

FFF Printing Fundamentals

Part 3: Dimensional Accuracy

Dimensional Accuracy

- Be careful modeling parts with exact dimensions.
- Why?
 - 3D prints are imperfect.
 - When assembled together, the fit is probably too tight.
- Where do these errors come from?
- And how do we fix it?



Dimensional Accuracy

- Sources of error include but are not limited to:
 - Parts that shrink or warp.
 - Inconsistent extrusions.
 - Racked gantry (skewed axes).
 - Shipping damage.
 - Incorrect belt tension.
 - Backlash, slop, or loose parts.
 - Moving parts that are worn down.
 - Sloppy slicer settings.
 - Thermal expansion (*discussed in Part 3*).
 - Vibrations (*discussed in Part 3*).

Modeling Calibration

- *(Printed size) = (Size) × (Scale factor) + (Constant offset).*
- Simplified model.
 - Assume fully functional motion system.
 - Apply to X and Y dimensions.
- Scale factor. Error changes with size.
 - Part shrinkage.
- Constant offset. Error is always the same.
 - Extrusion inconsistencies. Over and under-extruding.
 - Surface roughness. Additives like carbon fiber, wet filament, over-extruding.



(xkcd.com)

Fixing Scale Factor

- **Adjust part scaling in the slicer to fix scale factor.**
 - Scale XY size, or
 - Dedicated shrinkage compensation setting.
- Applies to
 - Large parts over 6 inches, 150 mm.
 - Filaments that shrink, like ABS.
- In my experience, safe to ignore for PLA, PETG, and small projects.

Fixing Constant Offset

- **Add clearance (air gap) between parts that touch to fix constant offset.**
- Very important for projects with assembled pieces!
- Best to add clearance while designing 3D objects.
 - Greatest flexibility. Independently adjust each surface.
- Or try adding negative XY offset in slicer.
 - Equally offsets all vertical surfaces.

Adding XY Clearance

- How much XY clearance? It depends ...
- Start with 0.25 mm. Offset each side by half, 0.125 mm.
- *Example:* Making a hole for a bolt.
 - Add 0.2 - 0.3 mm of clearance for a close fit.
- *Example:* Making a lid that easily slides into place.
 - Add 0.4 - 0.8 mm of clearance for a loose fit.

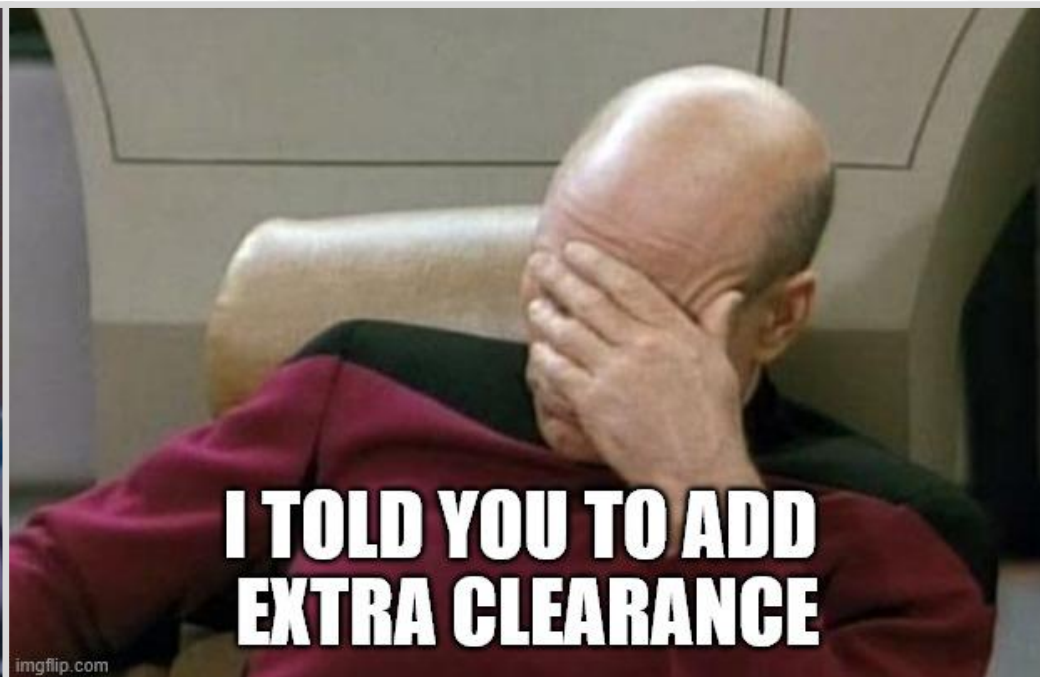
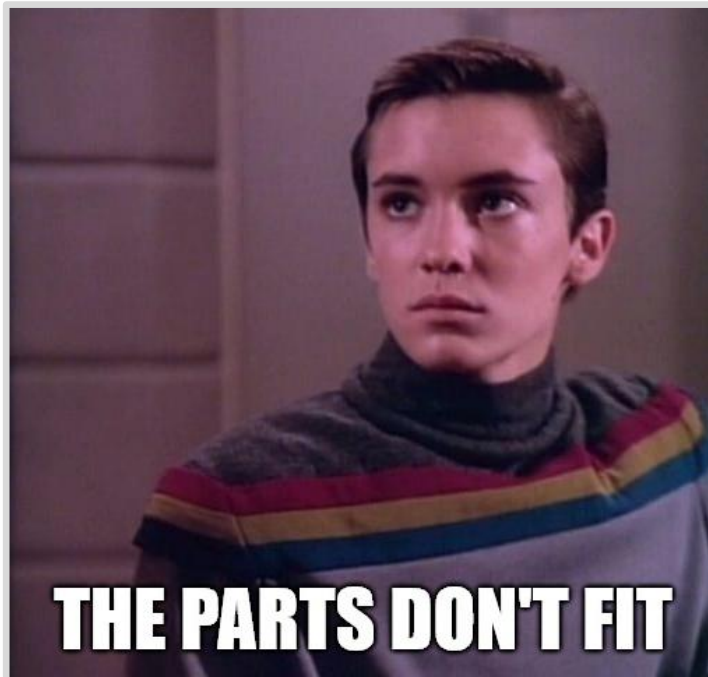
Adding Z Clearance

- How much Z clearance? It depends ...
- Start with one layer or 0.1 mm.
- Increase clearance in certain scenarios.
 - Sagging extrusions.
 - Rough bottom surface from supports.
 - High and low spots on print bed.
 - Tilted print bed.

Summary

If your project has assembled parts then ...

- You sometimes need scale factor.
- You always need constant offset.



Calibration Prints

- Calibration cubes.
 - 20 mm cube.
 - Labels for X and Y-axis.
 - Tells you constant offset for outside dimensions.
- Clearance test by 3DMakerNoob.
 - Print-in-place clearance/tolerance.
 - Alters the gap between pieces.
 - Feel which pieces move freely.



Calibration Prints

- Califlower by Vector3D.
 - Reasonably priced. Includes walkthrough and calculator.
 - Inner dimensions.
 - Outer dimensions.
 - Shrinkage.
 - Skew, when rectangles turn into parallelograms.
- Calibration video by CNC Kitchen.

